

STEP 2 — AUTOMATION ENGINE (GHL WORKFLOWS CLONE)

This is the **most important + most valuable part** of your system.

If you build this correctly → you can **sell this product easily**.

STEP 2 GOAL

Create a system where:

“When X happens → Do Y automatically”

Examples:

- Lead created → Send WhatsApp
 - Stage changed → Send Email
 - Wait 1 day → Send follow-up
 - If no reply → Send reminder
-

2.1 ARCHITECTURE OVERVIEW

We will NOT build everything from scratch.

👉 Use **n8n (automation engine)**

Final Flow:

Perfex CRM (Trigger Event)



Webhook → n8n



Workflow Logic (IF / WAIT / LOOP)



Actions (WhatsApp / Email / SMS)



Update back to CRM



2.2 INSTALL n8n (LOCAL + SERVER)



Option 1: Local (for development)

```
npm install -g n8n
n8n start
```

Access:

<http://localhost:5678>



Option 2: Production (RECOMMENDED)

Use Docker:

```
docker run -it --rm \
-p 5678:5678 \
-v ~/.n8n:/home/node/.n8n \
n8nio/n8n
```



Important:

- Set basic auth
 - Use HTTPS in production
-



2.3 CONNECT PERFEX → n8n (WEBHOOK)

You already created:

```
trigger_event('lead_created', $leadData);
```



In n8n:

Create workflow:

Node 1: Webhook

- Method: POST
- URL:

/webhook/lead_created

Test Payload

Perfex sends:

```
{
  "event": "lead_created",
  "data": {
    "id": 123,
    "name": "John Doe",
    "phone": "9999999999",
    "email": "test@gmail.com"
  }
}
```

2.4 WORKFLOW LOGIC DESIGN

Core Concept

Every workflow has:

Trigger → Condition → Action → Delay → Next Action

n8n Nodes You Will Use

Purpose	Node
Receive data	Webhook
Condition	IF
Delay	Wait

Send HTTP HTTP Request

Logic Function

2.5 EXAMPLE WORKFLOW (FULL)

Use Case:

When lead created → Send WhatsApp → Wait 1 day → Send Email

Nodes:

1. Webhook (Trigger)

Receives lead data

2. HTTP Request (Send WhatsApp)

Call your API:

POST <https://yourcrm.com/api/send-whatsapp>

Body:

```
{
  "lead_id": {{$json["data"]["id"]}},
  "message": "Hi {{$name}}, thanks for your inquiry!"
}
```

3. Wait Node

- Mode: Delay
 - Time: 1 day
-

4. HTTP Request (Send Email)

```
{  
  "lead_id": {{$json["data"]["id"]}},  
  "subject": "Follow up",  
  "message": "Just checking in..."  
}
```

2.6 CREATE API ENDPOINTS IN PERFEX

WhatsApp API Endpoint

```
public function send_whatsapp()  
{  
  $lead_id = $this->input->post('lead_id');  
  $message = $this->input->post('message');  
  
  $lead = $this->leads_model->get($lead_id);  
  
  // Call WhatsApp API (Meta/Twilio later)  
  
  // Save in conversations  
  $this->db->insert('tbl_conversations', [  
    'lead_id' => $lead_id,  
    'message' => $message,  
    'direction' => 'outbound',  
    'channel' => 'whatsapp'  
  ]);  
  
  // Log activity  
  log_activity($lead_id, 'whatsapp', $message);  
  
  echo json_encode(['status' => 'sent']);  
}
```

Email API Endpoint

```
public function send_email()  
{  
  $lead_id = $this->input->post('lead_id');  
  $subject = $this->input->post('subject');
```

```
$message = $this->input->post('message');  
  
$lead = $this->leads_model->get($lead_id);  
  
// Send email via SMTP  
  
log_activity($lead_id, 'email', $subject);  
  
echo json_encode(['status' => 'sent']);  
}
```

2.7 CONDITIONS (SMART AUTOMATION)

Example:

Only send message if lead status = NEW

n8n IF Node:

Condition:

```
{{ $json["data"]["status"] === "new" }}
```

More Conditions:

- Lead value > 5000
 - Stage = Qualified
 - Has phone number
-

2.8 DELAY & FOLLOW-UP SYSTEM

GHL Power Feature

You must support:

- Delay 5 min
- Delay 1 hour
- Delay 1 day
- Delay custom

In n8n:

Use **Wait Node**

2.9 LOOP / SEQUENCE SYSTEM

Example:

Send 5 follow-ups until reply

Logic:

1. Send message
 2. Wait 1 day
 3. Check reply
 4. If no reply → repeat
-

👉 You will check reply from:

tbl_conversations
direction = inbound

2.10 UPDATE BACK TO CRM

After automation:

Example:

- Mark lead as contacted
-

API:

```
public function update_lead_status()
{
    $lead_id = $this->input->post('lead_id');

    $this->db->where('id', $lead_id);
    $this->db->update('tblleads', [
        'status' => 'contacted'
    ]);
}
```



2.11 TESTING CHECKLIST

✓ Must Test:

- Webhook receiving data
 - WhatsApp API working (dummy OK)
 - Email sending
 - Delay working
 - IF condition working
 - Activity logging
 - Conversation saving
-



2.12 ADVANCED (OPTIONAL BUT POWERFUL)

Add Workflow UI inside Perfex (later)

Store workflows:

tbl_workflows

- id
 - name
 - trigger
 - json_data
-

👉 Later:

- Build drag-drop builder (like GHL)
-



STEP 2 OUTPUT

After this step, you will have:

- ✓ Automation engine (like GHL workflows)
 - ✓ Trigger → Action system
 - ✓ Delay + conditions
 - ✓ WhatsApp + Email automation
 - ✓ Scalable architecture
-



REALITY CHECK

At this point:

👉 You already built **70% of GoHighLevel power**

Most people NEVER reach here.